

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.3: Trigonometry — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Trigonometry
Driving Question	DRIVING QUESTION: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment — without ever measuring it directly?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Water Tank Problem — Launching the Anchoring Phenomenon	Two students (Amina at 10 m with a 38° angle of elevation, and Baraka at 20 m with a 22° angle of elevation) both sight the top of the same school water tank from different positions on the ground. Despite being at different distances and measuring different angles, they should calculate the same height — because the tank has only one height. We also observed that when we measured the ratio of the opposite side to the adjacent side across three right-angled triangles of different sizes but the same angles, the ratio stayed nearly the same.	A right-angled triangle is hidden inside the water tank problem: the vertical height of the tank is the OPPOSITE side, the horizontal distance from the observer to the base is the ADJACENT side, and the line of sight to the top of the tank is the HYPOTENUSE — all defined relative to the angle of elevation. The ratio of two sides of a right-angled triangle appears to depend only on the angle, not on the size of the triangle. The branch of mathematics that studies these ratios is called TRIGONOMETRY.	We cannot yet fully explain WHY the ratio stays constant, or HOW to use the 38° angle and 10 m distance to calculate the exact height of the tank. These questions are now posted on our Driving Question Board. We also cannot yet explain why a larger distance produces a smaller angle of elevation to the same point — we have observed it but need the trigonometric ratios (introduced in Lesson 2) to make sense of it mathematically.
Lesson 2 Sides of the Right-Angled Triangle — Naming What We See	Learners observed that in a right-angled triangle, the three sides have specific names — hypotenuse, opposite, and adjacent — and that the labels 'opposite' and 'adjacent' depend on which acute angle is chosen as the reference angle. They observed this by labelling paper triangles, re-labelling when the reference angle moved, and identifying right-angled shapes in real classroom objects including the water-tank diagram.	The hypotenuse is always the side opposite the right angle and is always the longest side — its name never changes. The 'opposite' side is the side directly facing the chosen acute angle θ , and the 'adjacent' side is the side next to θ that is not the hypotenuse. When θ moves to the other acute angle, opposite and adjacent swap. In the water-tank triangle: the line of sight = hypotenuse, the tank height = opposite, the ground distance = adjacent.	We now have a precise vocabulary to describe the water-tank triangle. The height of the tank that we cannot measure directly is the 'opposite' side, and the ground distance we can measure is the 'adjacent' side. In future lessons, we will use the ratio of these two sides — together with the angle of elevation — to calculate the exact height of the tank without climbing it.

Lesson 3 Deriving the Three Trigonometric Ratios: Sine, Cosine and Tangent	Learners measured the sides of right-angled triangles with angles of 30°, 45°, and 60°, computed the three side ratios (O/H, A/H, O/A) for each, and verified their values matched the calculator's sin, cos, and tan outputs.	The three trigonometric ratios — sine (opposite/hypotenuse), cosine (adjacent/hypotenuse), and tangent (opposite/adjacent) — are constant for a fixed angle regardless of the size of the triangle. The mnemonic SOH-CAH-TOA names and defines all three.	Using the tangent ratio and the horizontal distance from the base of the school water-tank tower, learners calculated the tank height from two different positions and obtained consistent results — confirming that trigonometric ratios make precise indirect measurement possible without ever touching or climbing the structure.
Lesson 4 The Constant Ratio — Discovering Sine, Cosine and Tangent Through Drawing	We drew three right-angled triangles — all with a 35° angle but with adjacent sides of 5 cm, 8 cm and 11 cm — and measured the opposite side and hypotenuse in each. We calculated opp÷hyp, adj÷hyp and opp÷adj for all three triangles and recorded the values in a table.	The three ratios (opp/hyp, adj/hyp, opp/adj) stayed almost exactly the same across all three triangles, even though the triangles were different sizes. The calculator confirmed the exact values: $\sin(35^\circ) \approx 0.574$, $\cos(35^\circ) \approx 0.819$, $\tan(35^\circ) \approx 0.700$. These constant ratios are called sine, cosine and tangent (SOH-CAH-TOA).	Because $\tan(\theta)$ is constant for a fixed angle, any observer — whether standing 6 m or 10 m from the base of the water tank — will calculate the same height using $\text{height} = \tan(\theta) \times \text{ground distance}$. The constancy of the ratio is the mathematical reason the water-tank measurement method works from any distance.
Lesson 5 Sine and Cosine of Complementary Angles	Learners computed sine and cosine values for complementary angle pairs (e.g., 30° & 60°, 40° & 50°, 45° & 45°) using mathematical tables or calculators, recording results in a structured table, and identified that $\sin \theta = \cos(90^\circ - \theta)$ across every row.	For any two complementary angles in a right-angled triangle, the sine of one angle equals the cosine of the other — because the same pair of sides is involved, just viewed from different angle positions. This means $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$.	In the water tank scenario, if Akinyi's angle of elevation (θ) and Baraka's angle of elevation ($90^\circ - \theta$) are complementary, then $\sin(\text{Akinyi's angle}) = \cos(\text{Baraka's angle})$ — confirming that the two students' measurements are internally consistent and that the same height can be recovered from either position using this shortcut.
Lesson 6 Sines and Cosines of Complementary Angles — The Hidden Shortcut in the Tank Problem	Learners completed a table of 7 complementary angle pairs (e.g., 35° & 55°, 45° & 45°) and computed sin and cos values, finding that the difference $\sin \theta - \cos(90^\circ - \theta) = 0$ for every pair, including the 38° elevation angle used in the water-tank calculation in Lesson 4.	In a right-angled triangle, the side that is 'opposite' to one acute angle is 'adjacent' to the other acute angle, because the two acute angles are complementary (they sum to 90°). This means $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ for all acute angles θ — so knowing one ratio immediately gives the other without re-measuring or recalculating.	The reason two students at different distances from the water tank — measuring different angles of elevation — calculate the same height is that their two elevation angles and the corresponding angle at the top of the triangle are complementary and consistent with the SAME fixed height. The complementary-angle identity shows that switching between sin and cos across the two acute angles of the right-angled tank triangle always produces equivalent expressions for the same physical length — confirming the consistency the phenomenon promised.
Lesson 7 Clinometer Construction & Field Calibration — Building the Measurement Tool	Learners constructed a clinometer from a protractor, string, plumb bob, and drinking straw; calibrated it against an indoor mark at a known height of 1.5 m and horizontal distance of 3 m (expected angle 26.6°); then went outdoors and measured the	Learners applied $\text{height} = \text{distance} \times \tan \theta$ to calculate the height of a real school structure from two different (distance, angle) pairs, confirmed that both calculations yielded nearly the same height, and explained in writing (CER format) why	Learners updated their cumulative explanatory model to include a labelled clinometer diagram, a right-triangle equation, and a consistency-check annotation showing that Amina at 10 m (42°) and Baraka at 15 m (30°) both arrive

	angle of elevation of a school structure from two different horizontal distances, recording all values in a results table.	the tangent ratio guarantees consistency — directly resolving the core puzzle of the anchoring phenomenon.	at the same height — and connected this method to how Kenyan surveyors and Mau Forest rangers determine heights without direct measurement.
Lesson 8 Final Explanation: Solving the Water-Tank Problem and Closing the Explanatory Loop	Learners used their self-built clinometers at two marked outdoor stations (8 m and 12 m from the school water-tank tower base) to measure and record the angle of elevation of the top of the tank from each position.	Learners learned that the tangent ratio ($\tan \theta = \text{opposite/adjacent} = \text{height/horizontal distance}$) can be rearranged to $h = d \times \tan \theta$, allowing the unknown height of the tank to be calculated precisely from a known horizontal distance and a measured angle of elevation — without ever climbing the tank or measuring its height directly. They also learned that two observers at different distances should calculate the same height, providing a built-in consistency check, and that small discrepancies arise from measurement uncertainty rather than errors in the mathematics.	Learners explained the complete anchoring phenomenon: the height of the school water-tank tower can be found by standing at a measured horizontal distance from its base, measuring the angle of elevation of its top with a clinometer, and applying $h = d \times \tan \theta$. This is the same method used by Kenyan surveyors in the Rift Valley highlands, KeNHA engineers designing Thika Superhighway overpasses, and foresters in the Mau Forest estimating tree heights for conservation — confirming that trigonometry is a powerful, universally applicable tool for finding unknown heights and distances without direct measurement.

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